

Instructor: Date:

Learner: Regional Branch:

Program: Center:

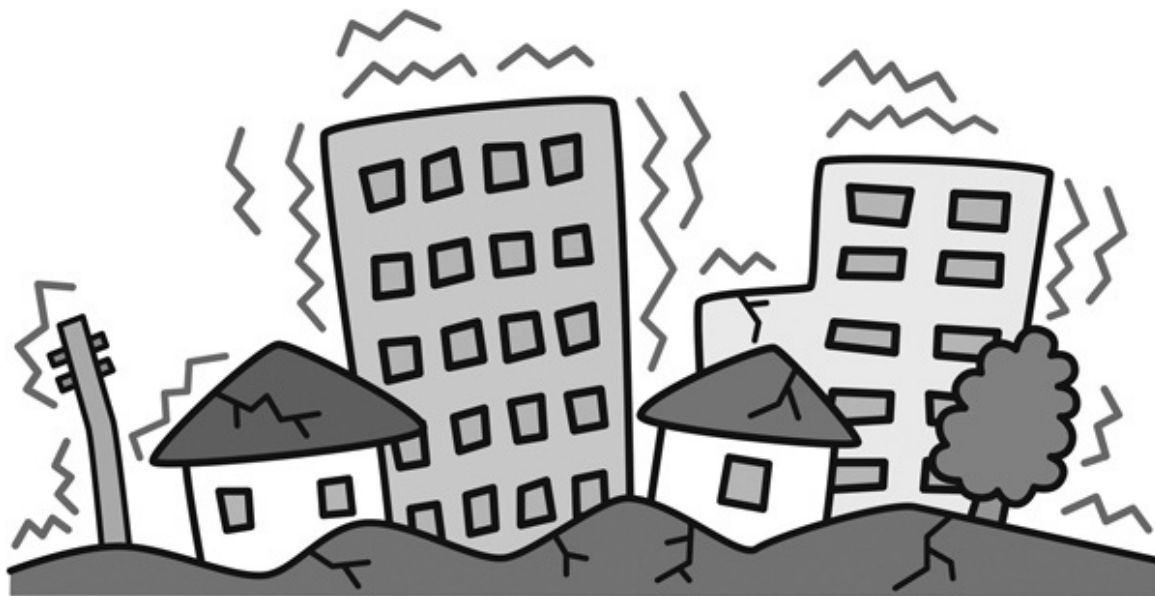
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Let's Start!

Was that an earthquake?!



Learning Outcome

Describe detailed facts about events or actions in work and social contexts, in written or spoken situations.

Ask your partner:

- What causes an earthquake?
- How would you feel about living where earthquakes are common?

Can Do Statements

- I can talk about geography, climate and environmental problems.
- I can talk about things that have happened.
- I can report facts about events or actions.





Let's Explore!

Small earthquakes* are happening around the world every day.

* "Small earthquakes" (2 or less on the Richter scale) happen more than one hundred times a day!



Instructions: Read the text aloud.

Scientists have said that, so far, earthquakes have been the most devastating of all natural disasters. Earthquakes happen where there are fault lines. A fault line is created by rocks that shift along cracks in the earth's crust. This means that the rocks beside each other are pulled apart. Many people have heard of the San Andreas Fault which is a long and wide fault in the state of California.

In countries such as Iran, Guatemala, Chile, and Japan massive earthquakes have often been recorded. Two of the biggest earthquakes that have ever taken place were in Alaska and China. These earthquakes measured over 8.5 on the Richter scale, which is the instrument used to measure earthquakes. Charles Richter, who was a seismologist, invented the scale in 1935.

On the Richter scale an earthquake that measures 2 or less is considered minor, but earthquakes that measure 7 and above are considered very dangerous and cause heavy damage.

It is important to predict when and where earthquakes will occur. Since the beginning of the twentieth century, Japan, the US and other countries have been actively assessing earthquake activity. In earthquake-prone areas they have established networks that look for warning signs that help predict the time when an earthquake is imminent.

There are other types of instruments that are used to detect activity in the earth's crust.

One of these instruments which is called a seismograph, from the Greek word *seismos* meaning earthquake, can detect movement in the earth thousands of miles away.

Seismologists have already been successful in their prediction of earthquakes in Russia and other parts of the world. However, it has been difficult to predict all earthquakes because new and more up-to-date detecting methods haven't been found yet. Early prediction may help areas prepare for an earthquake and limit its damage.





Let's Practice!

Earthquakes are calamitous*. What do you know about them?

*When something is "calamitous", it causes a lot of damage. This word comes from *calamity*.

Instructions: Complete the sentence by selecting the correct option. Write it down in the corresponding space.

1	2	3	4	5
up-to-date devastating best-known	Faults Seismographs Events	different imminent dangerous	crust signs damage	Levels Warnings Networks

- Earthquakes are believed to be the most _____ of all natural events.
- _____ are produced by the shifting of rocks along cracks which are in the earth's crust.
- Earthquakes that measure more than 7 on the Richter scale are considered very _____.
- Seismologists are in charge of studying the movements of the earth's _____.
- _____ of seismologists have been trying to predict the occurrence of earthquakes.



Let's Practice!

Earthquakes have frightened people since very ancient times ...

Instructions: Circle the word that is misspelled in each sentence. Write the correct word on the line.

- Two of the biggest earthquakes tha have ever taken place were in Alaska and China.

- Scientists have said that, so far, eartquakes have been the most devastating of all natural disasters. _____
- Other countries have been actively assessing earthquake activity. _____
- Seismologists have allready been succesful in their prediction of earthquakes in Russia. _____
- It has been difficult to predict all earthquakes because new and more up-to-date detecting metods haven't been found yet. _____





Extension Activity

Research what causes an earthquake and make a presentation.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Go online or to a library and find information on what causes earthquakes. You can also use diagrams or other visuals.
- Read and take notes on the information.
- Organize the information and rephrase it so it is not exactly the same as what you read.
- Give a 2 to 5 minute presentation on what causes an earthquake and include any diagrams or visuals that you found.

Self-Assessment

Yes No

I read about a natural disaster.

☐ ☐

I organize the information I have read.

☐ ☐

I write the information in my own words.

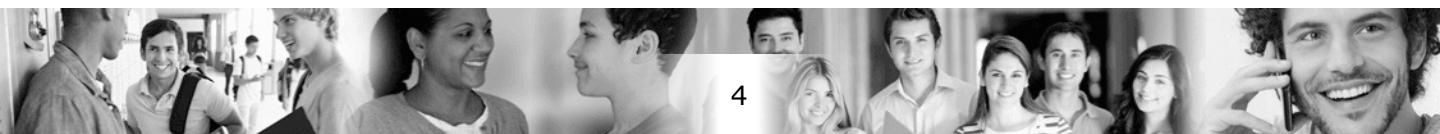
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I do a presentation with the information.

☐ ☐


Target Words

- earthquake** (noun): a sudden, violent movement of the earth's surface
e.g. Early prediction may help areas prepare for an earthquake and limit its damage.
- devastating** (adjective): causing a lot of damage and destruction
e.g. Scientists have said that, so far, earthquakes have been the most devastating of natural disasters.
- fault** (noun): crack in the earth
e.g. A fault is created by rocks that shift along cracks in the earth's crust.
- Richter scale** (noun): a system that uses a scale from 1 to 10 to measure the severity of an earthquake
e.g. On the Richter scale, an earthquake that measures 2 or less is considered minor.
- warning signs** (noun): early signals that something bad or dangerous may occur
e.g. The networks look for warning signs that help predict the time when an earthquake is imminent.
- earth's crust** (noun): the outermost solid layer of the earth
e.g. Many instruments are used to measure the change of the earth's crust.
- to detect** (verb): to discover something that is hidden or unclear
e.g. A seismograph can detect movement in the earth thousands of miles away.
- seismologist** (noun): a scientist who studies earthquakes, plate tectonics, and earth imaging
e.g. Seismologists have already been successful in their prediction of earthquakes in Russia and other parts of the world.
- up-to-date** (adjective): most recent; newest
e.g. It has been difficult to predict all earthquakes because new and more up-to-date detecting methods haven't been found yet.





Learning Points

The Present Perfect and Time Words

You use the Present Perfect tense to talk about past events or actions that began in the past but have a connection to the present.

There are certain "time" words that you usually use in sentences with the Present Perfect tense.

Look at the chart for some examples (note the position of the "time" word within each sentence):

Word	Use	Example
For	In sentences that describe something that has happened for a duration of time	<i>I have lived at my current address for seven years.</i>
Since	In sentences that describe something that has happened since a fixed period in time	<i>I have lived at my current address since 2008.</i>
Yet	In questions and negative statements	<i>I haven't taken out the trash yet.</i>
Already Often	In sentences that talk about actions that started in the past but have a present impact	<i>I have already taken out the trash.</i>
Still	In negative sentences	<i>I still haven't taken out the trash.</i>

Look at a few more examples from the text:

Examples:

- *In countries such as Iran, Guatemala, and Japan massive earthquakes have **often** been recorded.*
- *Two of the biggest earthquakes that have **ever** taken place were in Alaska and China.*
- ***Since** the beginning of the twentieth century, Japan, the US and other countries have been actively assessing earthquake activity.*
- *Seismologists have **already** had some success in predicting earthquakes in New York State.*
- *It has been difficult to predict all earthquakes because new and more up-to-date detecting methods haven't been found **yet**.*





Learning Points

Relative Clauses and Relative Pronouns

You use relative clauses to give extra information about someone or something. By combining sentences using relative clauses, your text becomes less repetitive and more concise.

The chart below outlines the main relative pronouns, and details how they can be used to form relative clauses.

Relative Pronoun	Refers to ...	Example
who	people	<i>I told you about the man who lives on the moon.</i>
which	animals or things	<i>Did you see the book which was on the table?</i>
whose	posessions	<i>I know a guy whose brother works at the circus.</i>
whom	people in the object position	<i>I ran into the doctor whom I met at my last check up.</i>
that	people, animals or things (replaces who or which)	<i>I told you about the man that lives on the moon. Did you see the book that was on the table?</i>

Look at a few more examples from the text:

Examples:

- *Two of the biggest earthquakes that have ever taken place were in Alaska and China.*
- *They have established networks that look for signs of an earthquake.*
- *Charles Richter, who was a seismologist, invented the scale in 1935.*
- *One of these instruments, which is called a seismograph, can detect movement in the earth thousands of miles away.*

